Dated 14/10/2026

HLC-Predictor: HLC-Predictor: A Henry's Law Constant Prediction Tool

Requirements:

- **HLC-Predictor: HLC-Predictor: A Henry's Law Constant Prediction Tool** is a Streamlit-based web predictive platform that only requires a web-browser.
- Requires **internet connection** for deployment or online hosting.

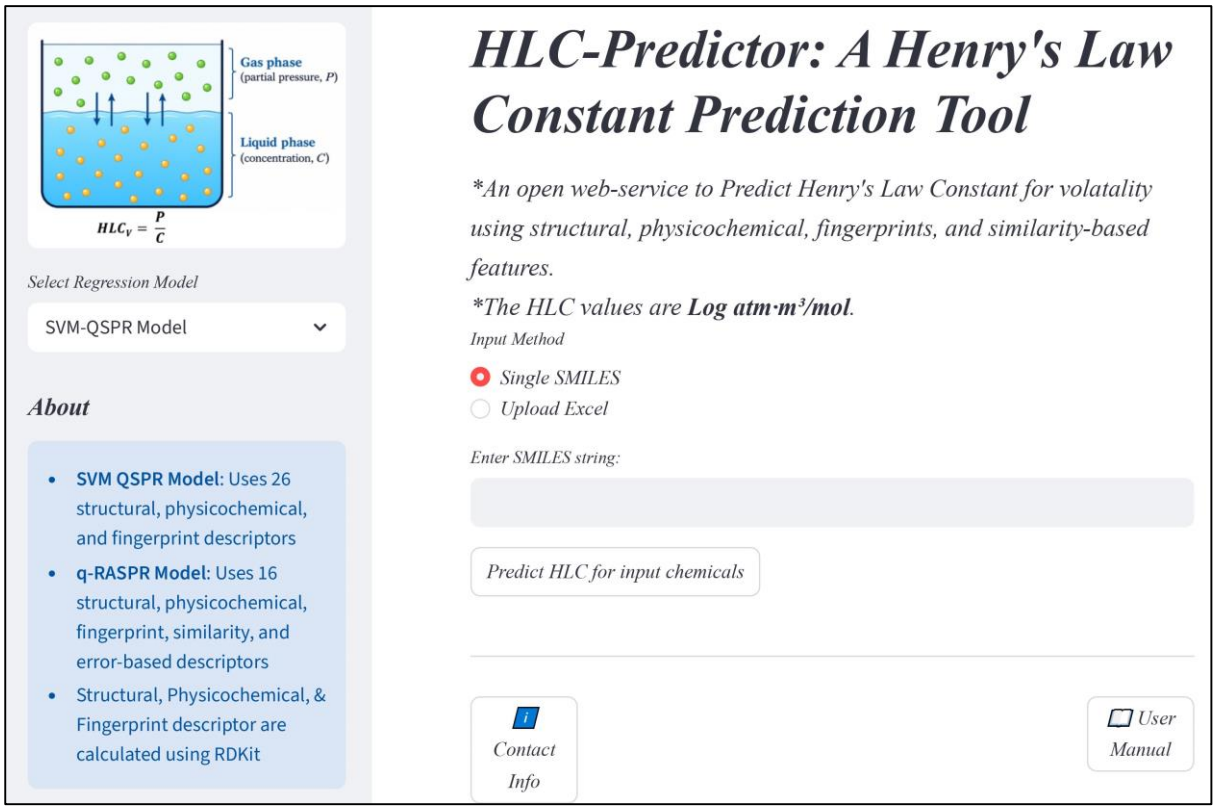


Figure 1: HLC-Predictor GUI

***Note:-** The user needs a valid SMILES string for the compound that is to be predicted.

Prediction Procedure:

1. To predict a single compound, the user can simply use ***Single SMILES*** as ***Input Method*** enter the SMILES string of the desired compound as input in the "*Enter SMILES string*" tab and then click the "*HLC*" button.

2. To predict a set of compounds, the user should prepare an Excel file (.xlsx format) containing the SMILES for each chemical in the first column of the first sheet with a proper heading. Then, the user must select "***Upload Excel***" as the input method and select the prepared file as input in the "*Upload*" tab by clicking on the button shown in **Figure 2**. Afterward, click on the "*Predict HLC for input chemicals*" button to obtain the predictions.

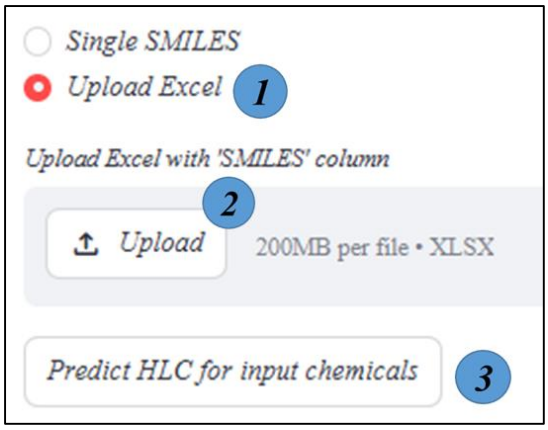


Figure 2: Steps to Upload SMILES file (.xlsx) for prediction

Input File:

- Must be in .xlsx format.
- A suitable heading in the first cell of the first column (**A1**).
- Valid SMILES in the first column from **Row 2**.

Table 1: Sample input file for the compounds needed to be predicted

	A	B	C
1	**SMILES**		
2	cccccc		
3	C(F)CCNC		
……..	……..		
……..	…….		

*The user can also refer to the "Sample.xlsx" file in the 'data' folder to prepare their input file.

View and Download Result:

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Once the prediction is complete, the results can be viewed directly within the interface. The output will include details such as SMILES (for the input compounds), predicted HLC (in atm.m^3/mol) values, and Applicability Domain (AD) status. Users also have the option to download the results in .csv format for single/batch process, which can be used for further analysis, recordkeeping, or reporting purposes.

Tool developed by



(GUI developed by **Shubham Kumar Pandey**)

For further queries, please contact

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